QŪNAR v0.8.1

USER GUIDE

Turkestan Region · v0.8.1

What QŪNAR does

QŪNAR supports scenario-based assessment of crop production, land and water use, energy demand and selected greenhouse-gas sources in the Turkestan region. Outputs are model estimates, not verified results of a real farm.

Getting started

Choose interface language, district, assessment season and task. Select Basic, Advanced or Research detail. Enter actual data where available; review every substituted reference value. Use the field boundary for a selected-crop assessment and the total cultivated area for farm-structure optimization.

How to interpret the output

Separate observed inputs, statistical references and modelled estimates. An improved contribution margin is not net profit. A feasible water balance does not establish an irrigation permit, and reduced emissions must be evaluated at matching activity boundaries.

Reports and reproducibility

Run Calculate in the selected mode, then prepare PDF, Word or Excel. My Reports collects chosen calculations. The integrated report consolidates repeated inputs at the end. Keep a record of reference year, source and model version.

1. Assess My Farm

Assess a selected crop plot, its irrigation, water and energy inputs, indicative margin and partial carbon footprint.

Water demand follows area and applicable norms; electricity, diesel and nitrogen contributions require a consistent selected-plot scope. Any irrigation-change margin is scenario-based.

Inputs

Enter the district and year, hectares and mode-specific quantities; confirm units and whether measurements refer to the selected plot or whole farm.

Method and output

QŪNAR combines statistical or user yields, agronomic norms, climate features and decision criteria. The active yield method may be ML or a historical fallback. Verify which method is actually used; the climate window for open-field ML is April–September, while resource inputs cover the full crop cycle.

Checks before decisions

Check missing-data warnings, reference substitutions, water feasibility, model confidence and scenario differences. Do not treat a ranking as guaranteed yield or profitability.

2. Check & Select a Crop

Evaluate one crop’s suitability, weather sensitivity, yield evidence and water norms.

Crop climate suitability is compared against historical district yields and selected method evidence; a fallback is used where calibrated ML support is lacking.

Inputs

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Method and output

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3. Recommend a Crop

Compare crops by multi-criteria ranking, climate response, water pressure and evidence reliability.

Scores combine normalized climate, water and evidence criteria; sensitivity to weights should be checked before changing crops.

Inputs

Enter the district and year, hectares and mode-specific quantities; confirm units and whether measurements refer to the selected plot or whole farm.

Method and output

QŪNAR combines statistical or user yields, agronomic norms, climate features and decision criteria. The active yield method may be ML or a historical fallback. Verify which method is actually used; the climate window for open-field ML is April–September, while resource inputs cover the full crop cycle.

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4. Optimize Farm Structure

Allocate total cropped area under land, crop-share, water and economic constraints; test stressed scenarios.

Allocation areas sum to the available cropped land; each crop has share caps and per-hectare water needs. Contribution margin = sum(area × indicative margin/ha).

Inputs

Enter the district and year, hectares and mode-specific quantities; confirm units and whether measurements refer to the selected plot or whole farm.

Method and output

QŪNAR combines statistical or user yields, agronomic norms, climate features and decision criteria. The active yield method may be ML or a historical fallback. Verify which method is actually used; the climate window for open-field ML is April–September, while resource inputs cover the full crop cycle.

Checks before decisions

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5. Assess Livestock Resources

Check pasture area, livestock numbers, feed balance, water demand and selected emissions.

Pasture load, available feed dry matter and demand are compared with stock size; watering and methane estimates depend on user activities and reference factors.

Inputs

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Method and output

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Checks before decisions

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6. Compare Crops and Livestock (beta)

Compare three mixed-farm configurations under explicit land and resource assumptions. Experimental mode; do not present it as field-validated.

Three scenarios compare crop/pasture structure and resource feasibility. No scenario is automatically superior if feed-chain, methane or land-conversion assumptions are unresolved.

Inputs

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Method and output

QŪNAR combines statistical or user yields, agronomic norms, climate features and decision criteria. The active yield method may be ML or a historical fallback. Verify which method is actually used; the climate window for open-field ML is April–September, while resource inputs cover the full crop cycle.

Checks before decisions

Check missing-data warnings, reference substitutions, water feasibility, model confidence and scenario differences. Do not treat a ranking as guaranteed yield or profitability.

METHODOLOGICAL NOTES

Contribution margin = crop sales revenue − applicable variable costs. It is not operating or net profit. Weighted field water = sum(area by crop × reference requirement). Optimisation maximizes the selected objective subject to crop-share, land and water constraints. QŪNAR may display Random Forest and comparison baselines; selection should follow evidence/validation, not model names alone. TOPSIS-like ranking depends on normalisation and indicator weights. FAO-56 water demand, irrigation-delivery norms and directly metered water are distinct concepts. CMIP6 deltas are scenario perturbations, not observed future forecasts. Partial CO_2e estimates depend on system boundaries and activity factors; missing activity data means comparison is unavailable.

Sources: QŪNAR v0.8.1 implementation; NASA POWER; FAO-56; CMIP6; Kazakhstan BNS; applicable irrigation norms. Confirm current source version before publication.